

Python for Data Science Practical Practical - VII Roll No. :47 Name: jyoti yadav Class : SY.IT Date of Assignment : 02/09/26

Write a Python program to create a scatter plot of Age and Height of 5 students.

```
import matplotlib.pyplot as plt

# Data for 5 students
ages = [18, 19, 21, 20, 22]
heights = [165, 172, 178, 160, 185] # in cm

plt.scatter(ages, heights, color='blue', marker='o', s=100)

# Formatting
plt.title('Student Age vs. Height')
plt.xlabel('Age (Years)')
plt.ylabel('Height (cm)')
plt.grid(True, linestyle='--', alpha=0.5)

plt.show()
```

b. Write a Python program to create a bar chart showing marks of 5 students.

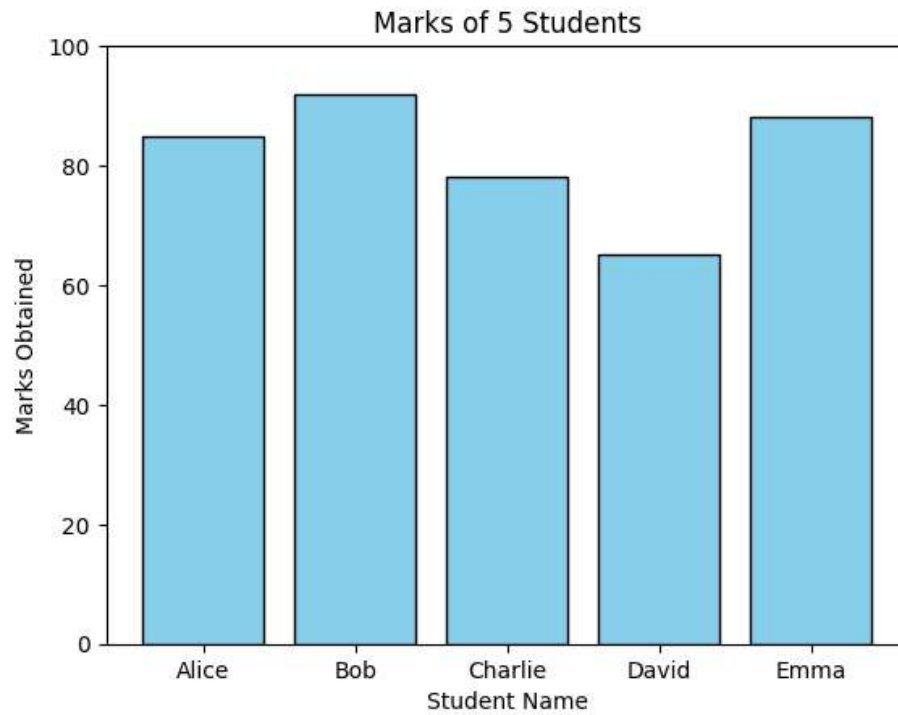
```
import matplotlib.pyplot as plt

# Data for 5 students
students = ['Alice', 'Bob', 'Charlie', 'David', 'Emma']
marks = [85, 92, 78, 65, 88]

plt.bar(students, marks, color='skyblue', edgecolor='black')

# Formatting
plt.title('Marks of 5 Students')
plt.xlabel('Student Name')
plt.ylabel('Marks Obtained')
plt.ylim(0, 100) # Set limit from 0 to 100 marks

plt.show()
```



c. Write a Python program to create a histogram showing the distribution of marks of 10 students.

```
import matplotlib.pyplot as plt

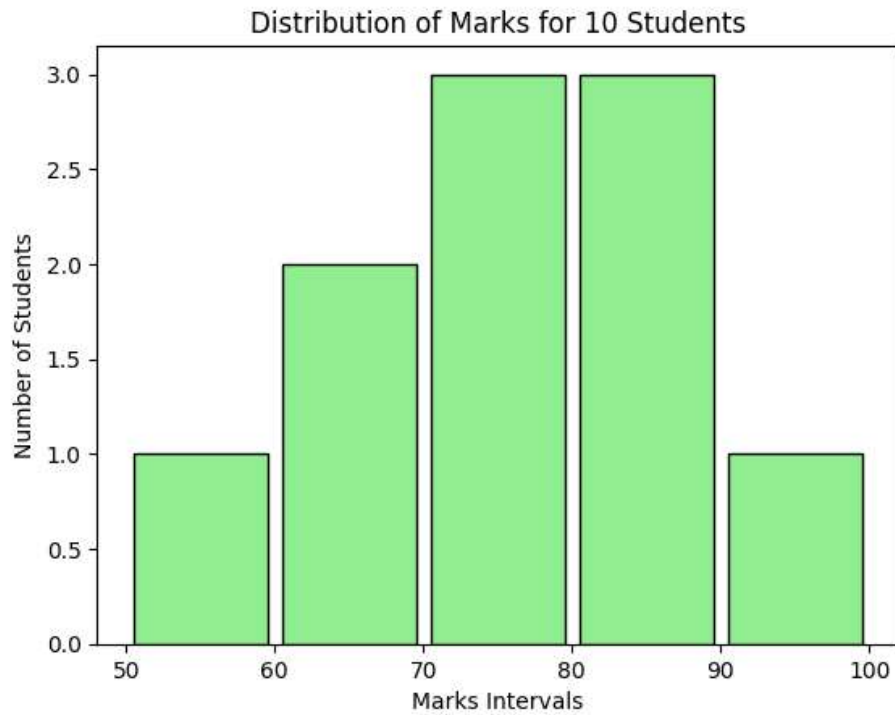
# Data for 10 students
marks = [55, 62, 67, 71, 74, 78, 82, 85, 89, 95]

# Defining bins for intervals (e.g., 50-60, 60-70, etc.)
bins = [50, 60, 70, 80, 90, 100]

plt.hist(marks, bins=bins, color='lightgreen', edgecolor='black', rwidth=0.9)

# Formatting
plt.title('Distribution of Marks for 10 Students')
plt.xlabel('Marks Intervals')
plt.ylabel('Number of Students')
plt.xticks(bins)

plt.show()
```



d. Write a Python program to create a pie chart showing the number of students in different courses.

```
import matplotlib.pyplot as plt

# Data categories
courses = ['Python', 'Java', 'Web Dev', 'Data Science']
student_count = [45, 30, 25, 40]
colors = ['gold', 'yellowgreen', 'lightcoral', 'lightskyblue']

# autopct displays percentages automatically on the slices
plt.pie(student_count, labels=courses, colors=colors, autopct='%1.1f%%', startangle=140)

# Formatting
plt.title('Student Distribution Across Courses')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

plt.show()
```

Student Distribution Across Courses

Data Science

e. Write a Python program to create a bar chart showing sales of 5 different products.

```
import matplotlib.pyplot as plt

# Data for 5 different products
products = ['Laptops', 'Smartphones', 'Headphones', 'Smartwatches', 'Tablets']
sales = [120, 250, 180, 95, 140] # Units sold or revenue

# Create the bar chart
plt.bar(products, sales, color='teal', edgecolor='black', width=0.6)

# Formatting the chart
plt.title('Sales Performance of 5 Products', fontsize=14, fontweight='bold')
plt.xlabel('Products', fontsize=12)
plt.ylabel('Units Sold', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.7) # Add horizontal gridlines for readability

# Display the chart
plt.show()
```

